Structure learning: does the hierarchy earn its depth? I

Study 7 shows the 3-level agent runs and federates end to end, but a deeper model is only warranted if the extra level carries information. We close the loop with a structure-learning test: given a trained hierarchy and one leaf observation, does Bayesian model reduction correctly decide whether the top meta-context level should be *kept* or *pruned*?

We reduce at the level granularity (:func:`fedference.bayesian_model_reduction.hierarchical_reduce`). For each non-leaf level we measure its **Bayesian surprise** $\mathrm{KL}(q_i \,\|\, \tilde{p}_i)$ — how far the leaf observation moves that level's belief q_i from its top-down prior $\tilde{p}_i$. A level whose belief the data never move carries no structure and is prunable; an informative level moves and is kept. This is an inference-derived divergence, not a model re-fit, so it cannot manufacture a difference the generative model does not contain.

The test is directional by construction. We build two 3-level worlds that differ *only* in the top level's conditioned priors: a **degenerate** world whose meta-context is non-gating (both meta-context states predict the same context distribution) and an **informative** world whose meta-context sharply distinguishes the two contexts. On the degenerate world the top level earns a Bayesian surprise of 0.000 nats and is flagged prunable (recovers the two-level structure: Yes); on the informative world the same level earns 0.328 nats and is kept (Yes); fig. 1 shows the per-level surprise for both worlds side by side. Because the two worlds share every other parameter, the opposite verdict is attributable to the meta-context's information alone — the reduction discovers the right depth rather than assuming it.

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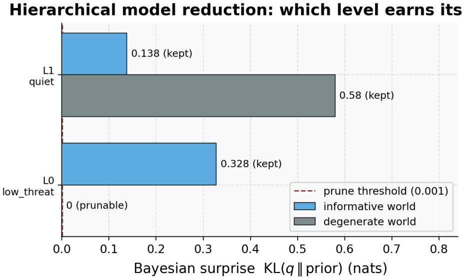

Figure 1: Source relation: original project BMR structure-learning diagnostic related to the mechanism in Friston et al. Fig. 9; estimand: per-level Bayesian surprise in nats and the resulting prune/keep decision; uncertainty: deterministic schematic worlds, so no resampling interval is shown. Per-level Bayesian surprise for the two 3-level worlds. y-axis: the non-leaf reduction targets, indexed top-down from the reduction routine as level 0 = the meta-context (the topmost non-leaf level, L3 in the location-first L1/L2/L3 convention used elsewhere) and level 1 = the context (L2); the leaf location level (L1) is never a reduction target. x-axis: Bayesian surprise $\mathrm{KL}(q \,\|\, \text{prior})$ in nats — the information the leaf observation added at that level. Blue bars: the informative world (top level kept). Grey bars: the degenerate world (top level prunable). The dashed red line is the prune threshold; a level whose surprise falls below it is structurally unnecessary. The degenerate meta-context (grey, level 0) sits at 0.000 nats and is pruned, recovering the two-level model, while the informative meta-context (blue, level 0) at 0.328 nats is retained; both worlds keep the context level (level 1). Deterministic schematic worlds (no resampling), so no error band is applicable.

This is the same Beta-function model-reduction machinery that drives the emergence study (sec. 7.4, eq. 16), lifted from pruning redundant *states* within one level to pruning a redundant *level* of the hierarchy — an honest, tested answer to "how deep should the generative model be?" that the data, not the modeler, decides.